

CAR CRASHES

Why America Has So Much Road Safety Research, But So Little Actual Safety

Why does all this research not translating into solid guidance that actually saves lives?



By Kea Wilson



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The Streetsblog Photoshop Desk

America is awash in taxpayer-sponsored road safety research, a new study finds — but that ocean of studies isn't always translating into solid guidance that actually saves lives.

In a sprawling new report from the National Academies of Sciences — a nongovernmental organization established under President Abraham Lincoln to advise the government on its scientific priorities — a committee of experts spent two years digging into why the American road safety crisis continues to escalate, even as publicly funded researchers release new insights into how to solve it almost every day.

Nearly 43,000 people lost their lives on U.S. roads in 2022 alone, even though those roads are designed and managed by transportation agencies that claim safety as their top priority, all of whom wield thick design manuals supposedly backed by decades of data.

“When you walk into a state Department of Transportation, the sign on the door usually says, ‘Safety is our most important objective,’ and [you’d hope] that would actually play out in their day-to-day operations,” said Joseph Schofer, a professor emeritus at Northwestern and the chair of the committee. “But I would say it does not seem to be what is most important. It seems to be that in many state departments of transportation, it’s more about maintaining the construction program, about employment and spending money. Or it’s about throughput: pushing more traffic through a fixed network.”

Schofer and his colleagues identified five key factors that have shaped that paradoxical status quo — and recommendations for how not just gather the best research possible, but actually implement it on our roads.

First, they say that there are simply too many road safety researchers out there working at once without a unified vision for which crash-cutting strategies deserve their urgent attention. Establishing a “National Road Safety Research Agenda,” the team argued, could help focus their energies on what matters most.

“What you have out there is, at minimum, three parallel research programs: one sponsored by the Federal Highway Administration, one sponsored by AASHTO, one sponsored the National Highway Traffic Safety Administration. ...There’s no integrated agenda, and we need one,” Schofer added.

Second, the committee says that many common crash countermeasures aren't rigorously re-evaluated as our transportation landscape evolves — and that even some legally-required solutions in our road design manuals “are not supported by research having a strong empirical basis.”

A full three pages of the report, for instance, is devoted to unpacking the notorious “85th percentile rule,” which recommends that communities set speed limits based on the velocity of the 85th-fastest driver on the road — even though the study that generated that method dates back to an analysis of rural road networks in the 1930s, and is “dependent on driver behavior, not analyses of safety.”

Even the most current road safety research, of course, isn't worth much if it's applied in the wrong context — and since every street is its own unique puzzle, even the best-intentioned practitioner might struggle to identify when a bump out is better than a speed hump.

“The technical guidance that traffic engineers and transportation planners use is grown by accretion,” explained Schofer. “So there are multiple sources of guidance, and even when a highly motivated planner or engineer sits down to try to address a problem, she or he will very commonly find that if you look at three sources of guidance, they'll suggest three different things. And so there's a need to integrate, there's a need to coordinate, and then needs to fill that gap.

“If you think about Complete Streets, it's 10, 12, 14 different things you can do in a in any one place,” he continued. “The kind of research that we're looking for is high quality research that says: ‘if you do these things, here's what the outcome's going to be.’ And we don't always have that.”

Schofer and his colleagues acknowledge that road design is enough of an art that it might not always been possible to offer one quick, scientific analysis of which design solutions will save the most lives. At the very least, though, he and his colleagues say that the mess of design guidelines transportation pros look to now could be integrated into a single “one-stop source of practitioner guidance, that is designed to be more accessible and easier to apply.”

Actually applying that guidance, of course, will depend on transportation engineers who understand what safety is all about — something which American transportation engineering programs aren't exactly famous for producing.

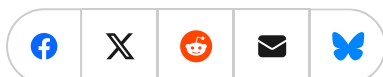
“What they’re studying, but for the most part, is traffic operations,” added Schofer. “They learn about how to maximize throughput. They learn about to make a road network most efficient. ... The committee came to the conclusion that it would be really helpful if there was some kind of nugget of road safety training that’s brought to people getting undergraduate degrees in engineering, to give them that core knowledge.”

Finally, the committee acknowledged that these structural changes will take serious resources — and US DOT isn’t currently set up to provide them. That’s why they say the agency should stand up a “National Road Safety Research Center with a focused mission to promote road safety research and its translation to practice.”

Even if all these recommendations are implemented — and considering that this isn't the Academies' first time exploring this issue, there’s no guarantee — Schofer acknowledges that we won’t end the national road safety crisis overnight. But there’s no more time to waste before America tackles these foundational problems.

“What the Safe Systems approach suggests, and what our committee fully adopted, was the notion that safety really ought to be in everything you do — and it ought to be a *main* thing that you do,” he added. “Killing 43,000 people a year is not the cost of doing business; it isn’t ethically responsible to walk away from that and say, ‘Well, that’s what it costs to have a great and capacious highway system.’”

A previous version of this article misspelled Joseph Schofer’s last name.





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Kea Wilson is Senior Editor for Streetsblog USA. She has more than a dozen years experience as a writer telling emotional, urgent and actionable stories that motivate average Americans to get involved in making their cities better places. She is also a novelist, cyclist, and affordable housing advocate. She lives in St. Louis, MO. For tips, submissions, and general questions, reach out to her at kea@streetsblog.org, or on Bluesky [@keawilson.bsky.social](https://bsky.app/profile/keawilson.bsky.social).

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